

Public Economics and Policy

Lecture #2

Theory and application of public finance (I)

August 11, 2026

1

1

Learning objectives

- ☐ Understand economic interactions between sectors using a circular flow diagram
- ☐ Explain the role of government through a multiplier effect
- ☐ Calculate a fiscal multiplier in a given scenario
- ☐ Understand how the population's age structure affects a fiscal multiplier

2

2

Outline

- ❑ The interactions between households, firms, government and foreign trade in a circular flow diagram
 - ❑ Two-sector model
 - ❑ Three-sector model
 - ❑ An open-economy model
- ❑ Aggregate demand in an economy
 - ❑ The fiscal multiplier
 - ❑ The fiscal multiplier in a closed economy without taxation
 - ❑ The fiscal multiplier in a closed economy with taxation
 - ❑ The balanced-budget multiplier in a closed economy
- ❑ The Paradox of Thrift
- ❑ Population's age structure and its impacts on a fiscal multiplier
- ❑ Conclusion

3

3

Reading

- ❑ Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)
- ❑ Thomas R. Michl. (2015). *Macroeconomic Theory: A Short Course*. Routledge. (Chapter 3)
- ❑ Krish Bhaskar, & David F. Murray. (2015). *Macroeconomic Systems*. Routledge. (Chapter 4, 5)

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Review question

- ☐ Does a government aim at maximizing a profit?
- ☐ If not, what are the objectives of government function?

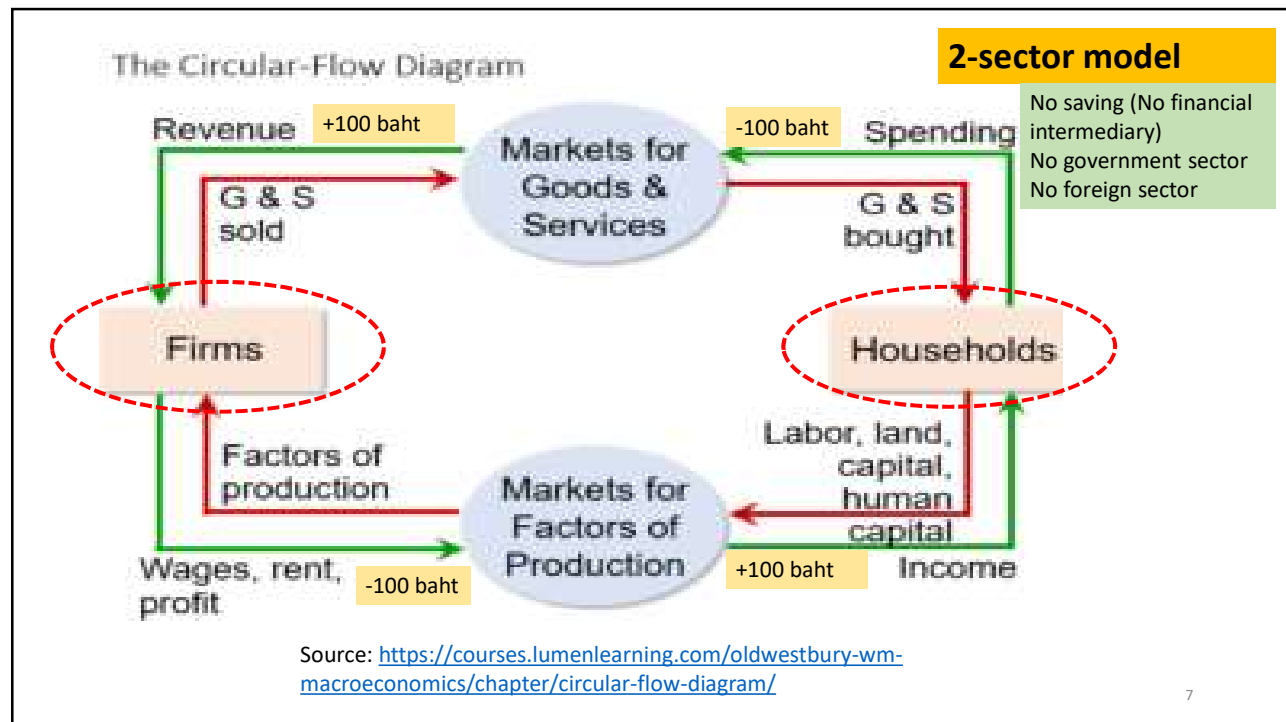
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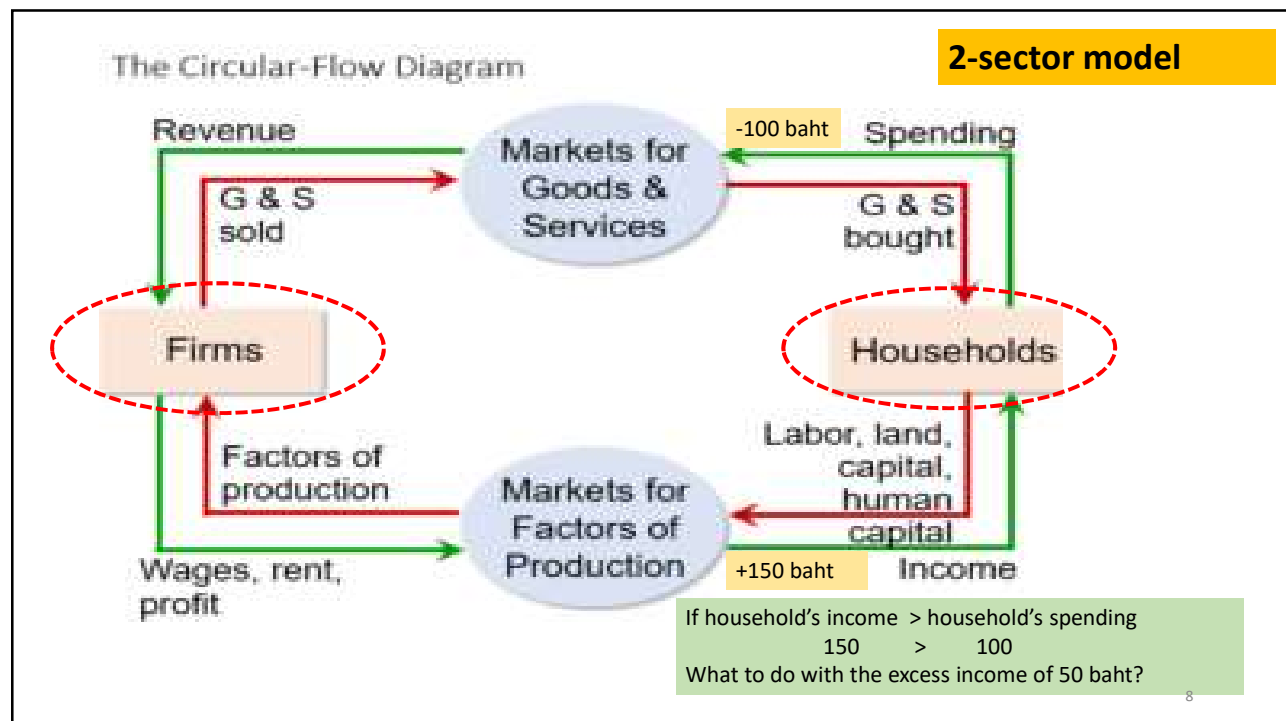
**The interactions between households,
firms, government and foreign trade
in a circular flow diagram**

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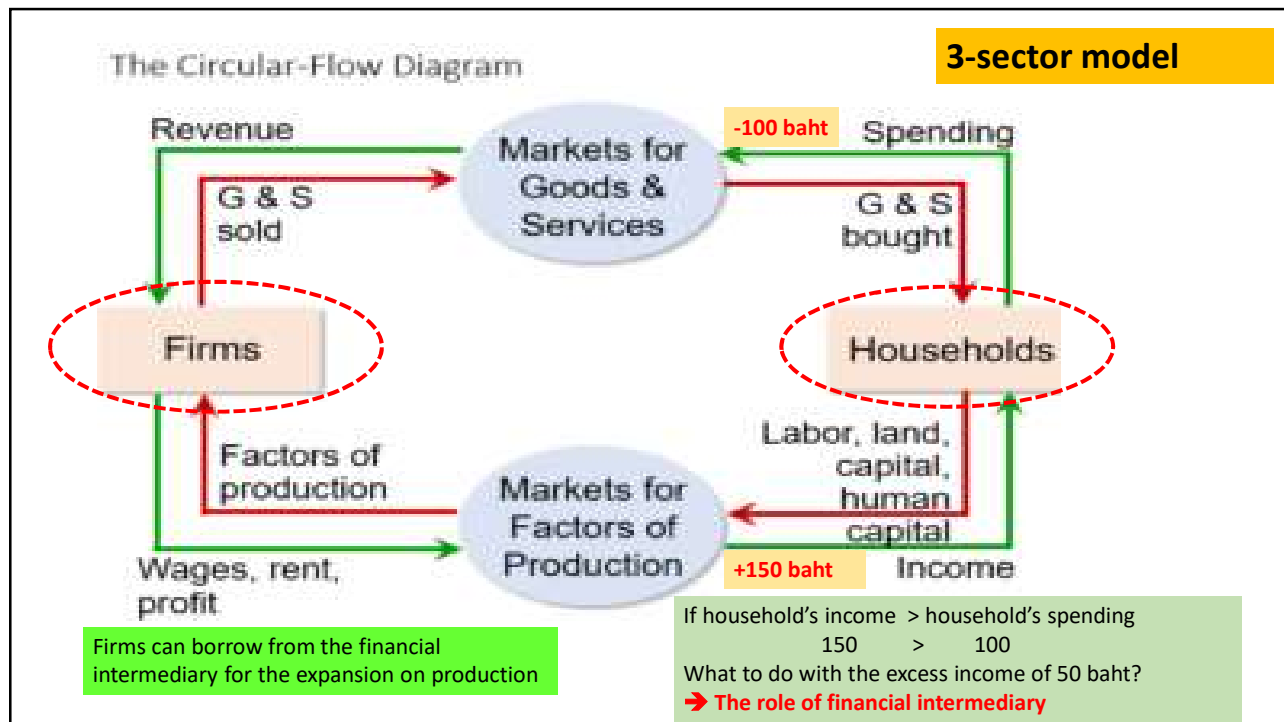
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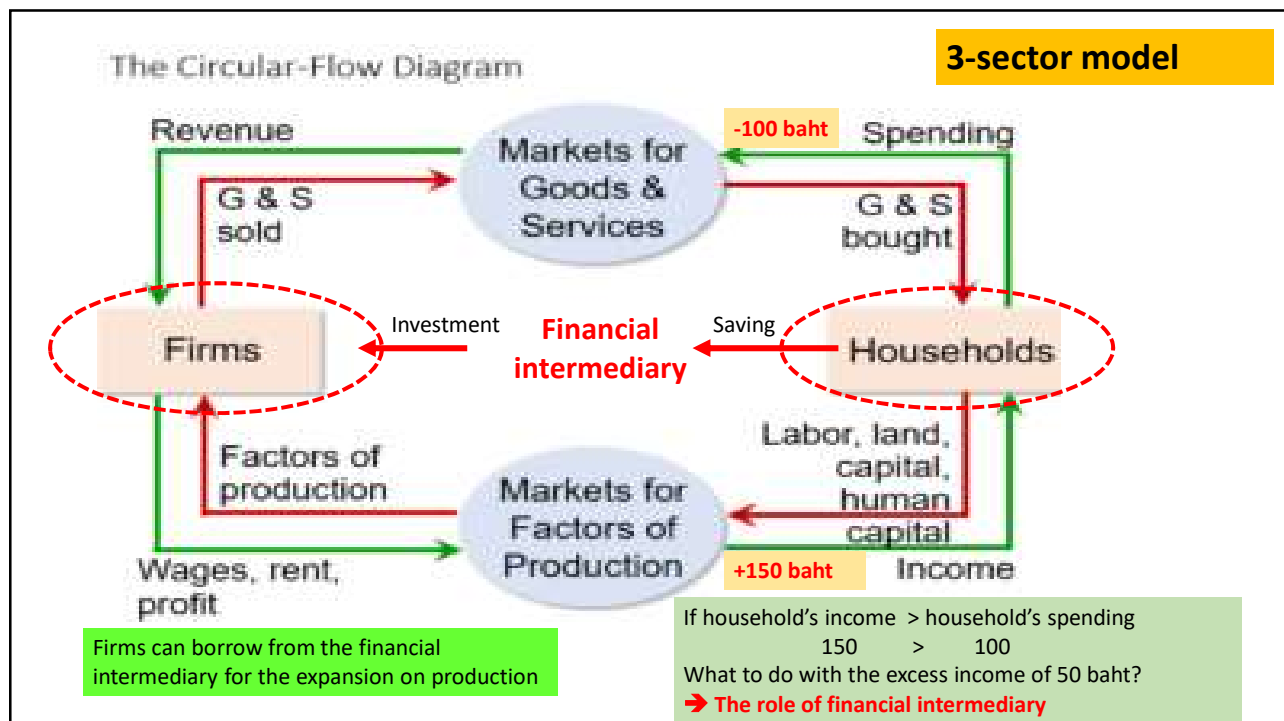
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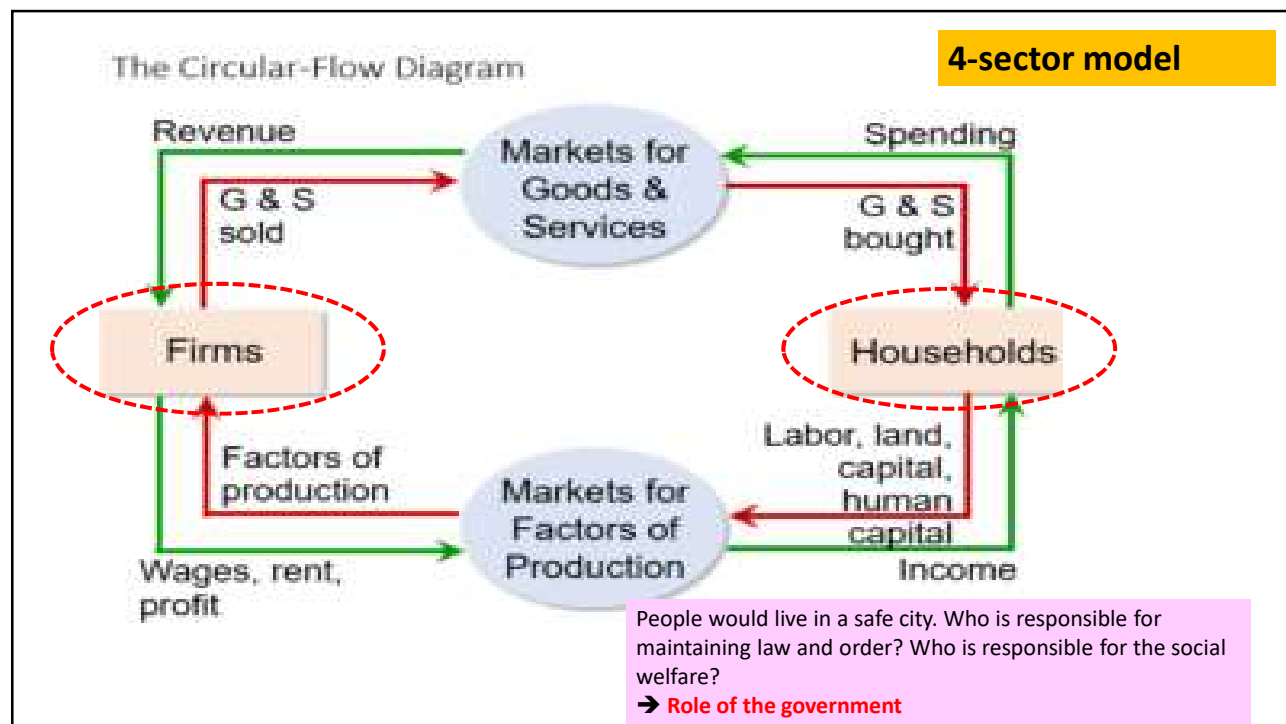
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Recall: Government Functions

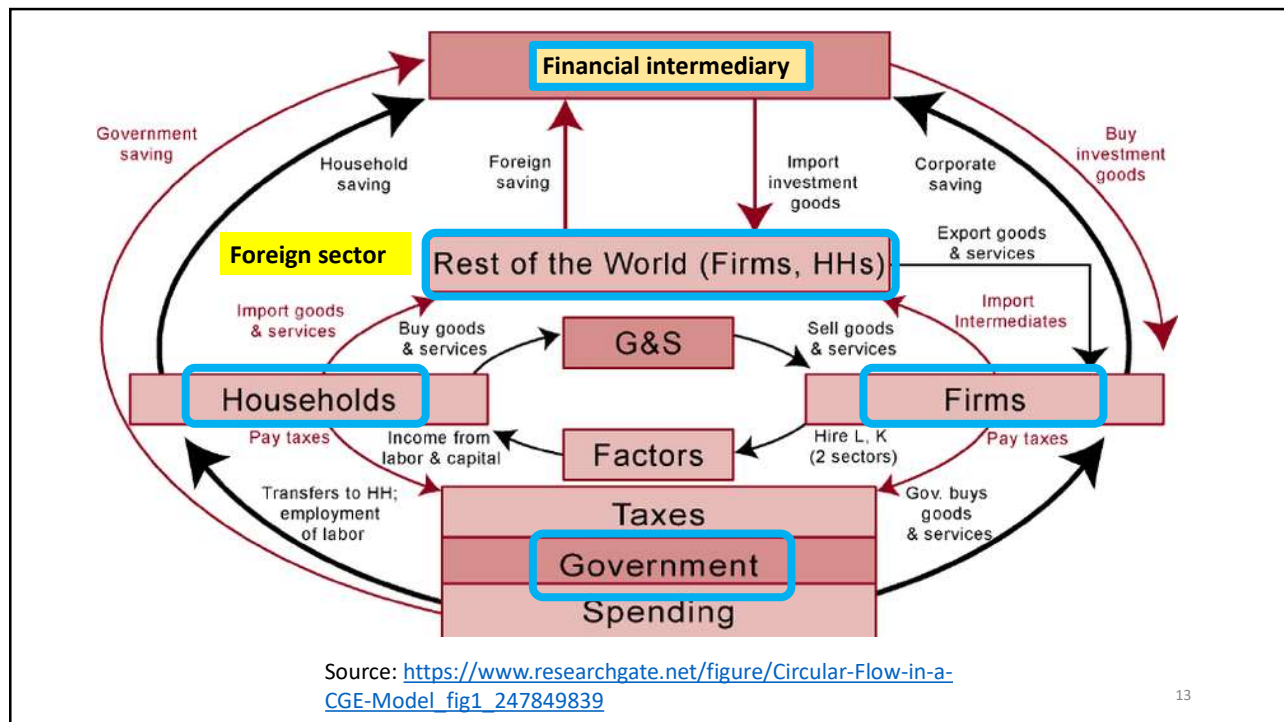
Resource Allocation

❑ Public goods

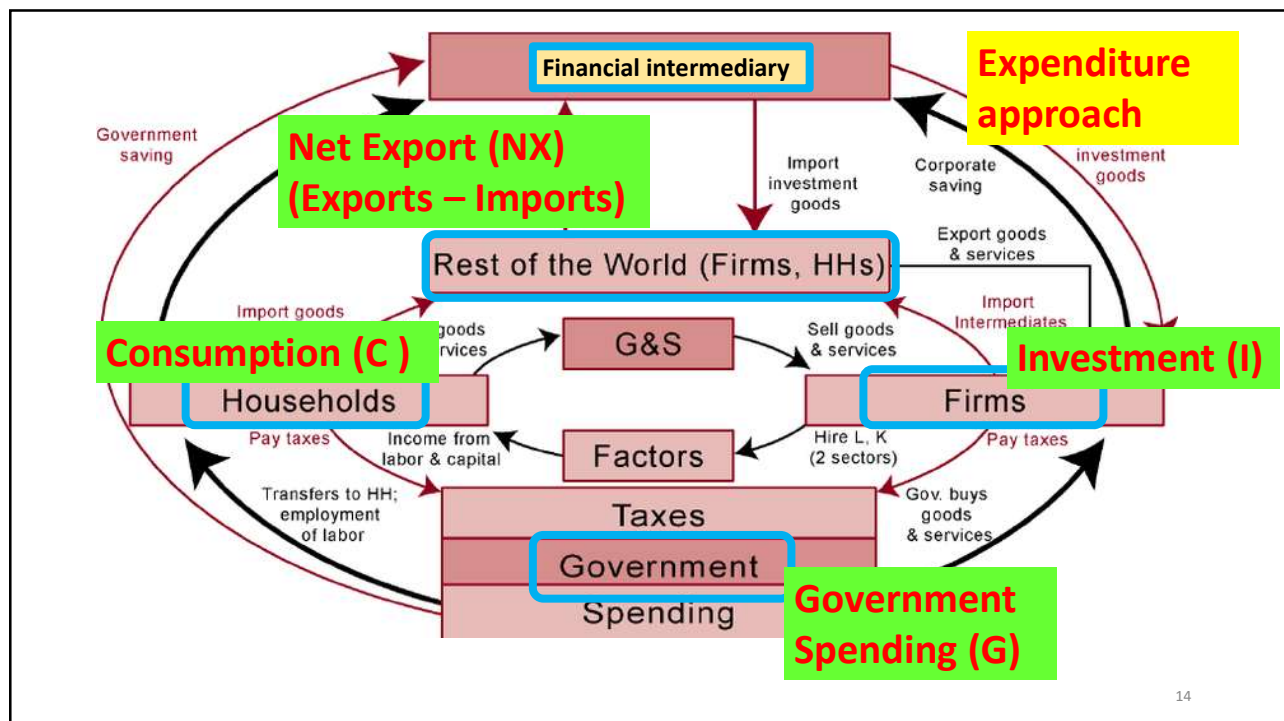
- ❑ These public goods are not well provided in the private sector. (Because the benefit of public goods has positive externalities.)
- ❑ If agents voluntarily provide these goods, others can consume the benefit without paying for the burden. → Lead to the ***"Free rider problem"***

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How does a government spending affect the economy?

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Aggregate Demand (AD)

Expenditure approach

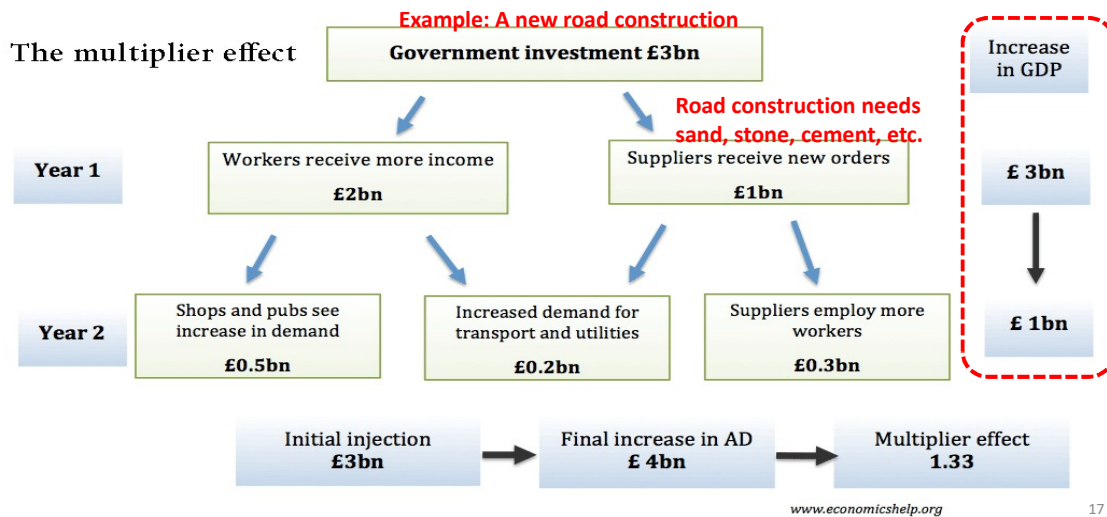
- ☐ In a closed economy: $AD = C + I$
- ☐ In a closed economy with government: $AD = C + I + G$
- ☐ In an open economy: $AD = C + I + G + NX$

C – Consumption expenditure
 I – Investment expenditure
 G – Government expenditure
 NX – Net exports or (Exports – Imports)

16

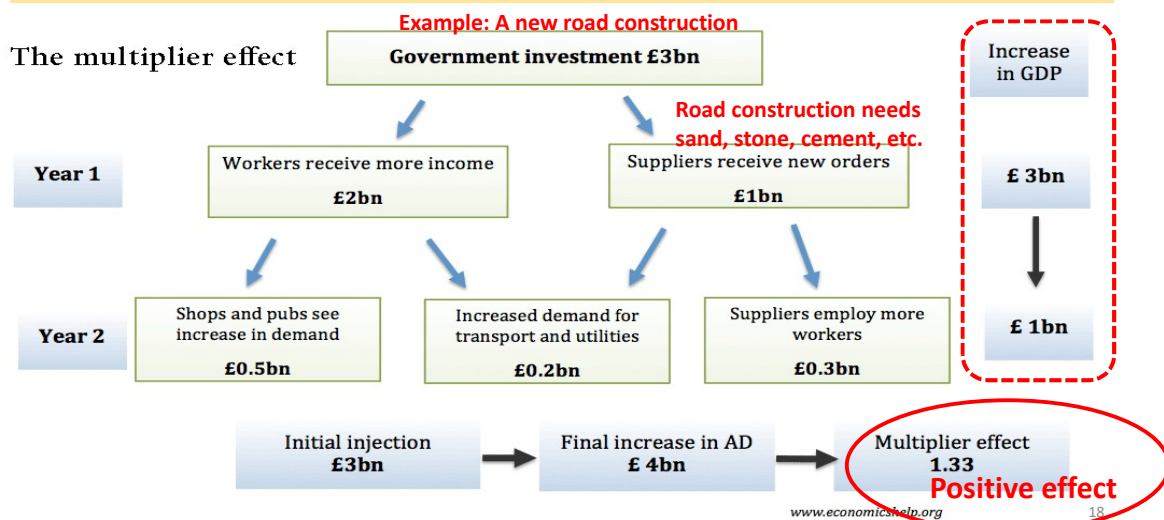
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How does a change in the government expenditure affect the aggregate demand in an economy?



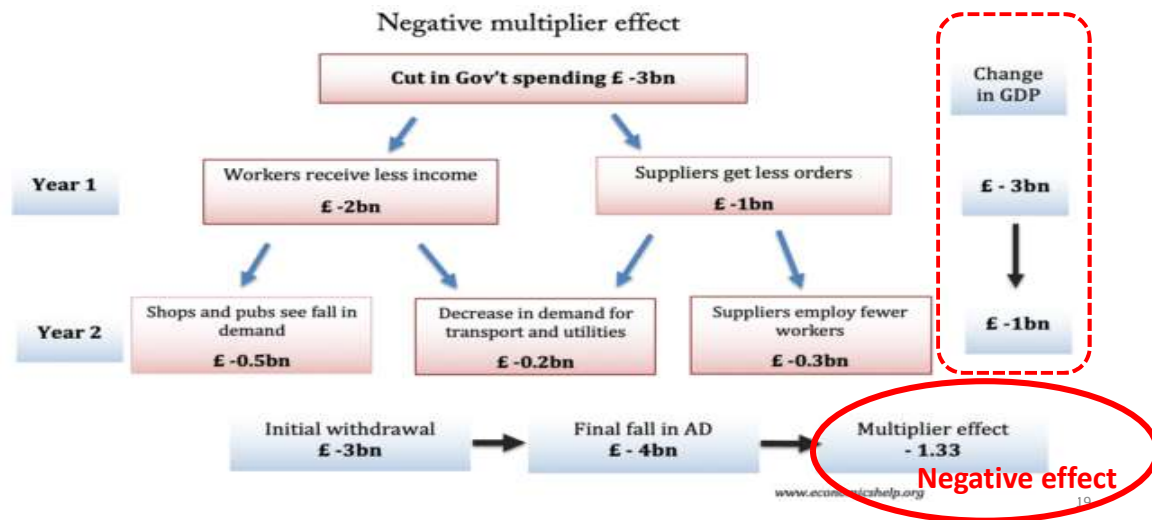
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How does a change in the government expenditure affect the aggregate demand in an economy?



18

In opposite, what if the government cuts in expenditure?



19

From the circular flow diagram, we have seen that ...

- ❑ National income increases when government spending rises → Expansionary fiscal policy (including lowering tax rate and increase in social benefits)
- ❑ In opposite, national income decreases when the government cuts in spending → Contractionary fiscal policy (including a rise in tax rate)
- ❑ *Question: How do we calculate a fiscal multiplier?*

20

20

Conceptual framework: The Simple Keynesian Model

21

21

The 45-degree Model

1.1 The 45-degree Model

❑ **Key point: the effect of the size of the multiplier on GDP**

❑ Review 2-sector model, 3-sector model, 3-sector model with government, and open economy

❑ The equilibrium condition for the goods market is given as

$$Y = C + I + G \quad (2.1) \text{ (closed economy)}$$

$$C = cY \quad ; \text{ c is MPC ,and } 0 < \text{MPC} < 1 \quad (2.2)$$

❑ Substituting Eq. (2.2) into Eq. (2.1) and eliminating C, we have

$$Y = cY + I + G \quad (2.3)$$

❑ Note:

1) $Y = C + I + G + \text{NX}$ --- open economy , NX (Net export) is Export minus Import

2) $\text{MPC} + \text{MPS} = 1$, (MPC is Marginal Propensity to Consume) (MPS is Marginal Propensity to Save)

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

22

22

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23

23

The Fiscal Multiplier

1.2 The Fiscal Multiplier

From Eq. (2.3), if we have the difference and consider $\Delta I = 0$, we obtain

$$\Delta Y = c\Delta Y + \Delta G$$

The fiscal multiplier ($\Delta Y/\Delta G$) is

$$\frac{\Delta Y}{\Delta G} = \frac{1}{(1 - c)}$$

❑ Numerical example

❑ Calculation and interpretation

❑ An example of open economy model

Source: Ihuri, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

24

24

The Multiplier with Tax Rate

1.3 The Multiplier with Tax Rate

Suppose the tax function is given as $T = tY$ (2.5)

Then, the consumption function is

The size of the fiscal multiplier with the tax rate is

$$\frac{\Delta Y}{\Delta G} = \frac{1}{[1 - c(1 - t)]}$$

Calculation and compare the result to the economy without tax

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

25

25

The Balanced-Budget Multiplier

1.4 The Balanced-Budget Multiplier in a closed economy

❑ A balanced-budget constraint

$$\Delta T = \Delta G$$

❑ Find the balanced-budget multiplier and interpret.

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 2)

26

26

The Paradox of Thrift

- ☐ What if everyone saves more of his money? What would happen to the GDP of that country?
- ☐ Discuss ...

27

27

The Paradox of Thrift

- ☐ How would an increase in private saving affect the level of GDP?
- ☐ Most likely, increased saving implies reduced consumer demand, which causes firms to cut back on production.
- ☐ Thus, an increase in saving actually causes a lower level of economic activity reduce the level of output.

28

28

Conclusion

- ❑ The circular flow model illustrates how income, expenditure, products, and inputs circulate through an economy. It shows the interactions between households, businesses, government, and foreign trade.
- ❑ The multiplier effect measures the impact of changes in government spending or taxation on economic output (GDP). It quantifies how an initial injection of government spending leads to a larger increase in national income.
- ❑ The fiscal multiplier expresses how much national income changes relative to changes in government spending. It depends on the marginal propensity to consume (MPC).

29

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30